
Skills

Design Distributed Systems, Data ETL, Performance Optimization, Machine Learning
Languages C/C++, Python, Java, Go, SQL, R, Bash

Experience

2018 Apr – **Senior Software Engineer**, *Google - Search*.

- Present Envisioned and lead the team implementing ML training and serving ecosystem for Search's News and Content Discovery.
- o Designed and lead the implementation of unified and standardized ML infrastructure for Discover, Notifications, Suggest and user modeling teams featuring:
 - signal retrieval and rich inference serving platform (C++),
 - zero training/serving skew near real time training data generation system, reducing the feature engineering work by 70% across the org (Go, C++),
 - reinforcement learning support (in collaboration with DeepMind),
 - dynamic triggering rate optimization based on recent and long-term user activity pattern (C++).
 - o Scaled an existing inference serving infrastructure to serve 24x more inferences using the same resources.
 - o Restructured Discover Search training process to support training on TPU to achieve 40x speedup (Python).
 - o Launched a flexible accelerator resources acquisition and scheduling solution to improve GPU and TPU utilization by 140% (in collaboration with DeepMind).
 - o Created user opt-out detection system proactively stopping bad experiments (C++).

2016 Apr – **Software Engineer**, *Google - YouTube*.

- 2018 Apr Built storage abstraction microservices framework with no-downtime migration support (C++).
- o Designed storage management framework allowing the user to create a full set of standard RESTful APIs for a defined resource in several hours regardless of the storage type (C++).
 - o Performed video metadata migration between Vitess and Spanner with no downtime, data loss, or client latency increase (C++). Improved YouTube Public Data API performance, usability, and security (Python).
 - o Performed data access patterns analysis to identify convenience methods to be added to the API, improving performance and simplifying client side code (Python).
 - o Built traffic analysis dashboards utilizing Fourier transform to visualize client requests in frequency domain and allow for simple in-service fraud detection (Python).
 - o Enforced end user credential checking to prevent private data leaks (Python, C++).

2015 **Fellow**, *Insight Data Science*.

- Sep–Oct Built CityGravity.info, a trip schedule optimization app to see the greatest number of sites while staying under budget during a visit to a new city.
- o Supplemented Expedia static database with data from Google Directions and EAN APIs and crime data scraped from the spotcrime.com JavaScript elements; processed raw JSON and XML outputs into a regularly updated MySQL database
 - o Used DBSCAN and hierarchical clustering to achieve optimal attractions' cluster size; interfaced Fortran subroutine with Python for faster distance matrix computation
 - o Utilized MySQL to store and efficiently retrieve the pre-computed data from multiple tables
 - o Implemented Python algorithm to recommend an accommodation minimizing travel time between the locations of interest
 - o Deployed an interactive front end using Flask, Google Maps, Javascript, jQuery and AWS

2010 Aug – **Graduate Researcher**, *University of Arizona, Tucson, AZ*.

- 2016 Jan Developed Fortran code modeling the molecular system total wave function as an optimal combination of explicitly correlated Gaussian functions. Generalized the approach to be applied to exotic atoms and molecules and extended it to excited states. The results are published in over a dozen peer-reviewed papers and presented on multiple physics and supercomputing conferences.
- o Improved the code scalability and memory requirements by combining MPI and OpenMP frameworks
 - o Designed the stochastic basis function selection algorithm, which boosted the calculation accuracy by 3 orders of magnitude for electronic and 7 orders for muonic systems at the cost of insignificant number of additional operations

2010 **Intern**, *Joint Institute of Nuclear Research, Dubna, Russia*.

- Jan–May Contributed to muon detector for CMS software development by assessing signal processing code and vectorizing it

Education

2010–2015 **Ph.D.**, *University of Arizona, Tucson, AZ – Physics*.

2005–2010 **B.S. (Hons.)**, *Saratov State University, Saratov, Russia – Physics*.